

# Geometry, Entropy, and Operator Learning with Tunable Kernels

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July 12, 2025

## Abstract

We study a two-dial family of kernels

$$k_{\theta,\sigma}(y \mid x) \propto w_{\theta}(\|x - y\|) g_{\sigma}(y - x),$$

whose product form covers any radial weight  $w_{\theta}$  and centred residual density  $g_{\sigma}$  that satisfy minimal smoothness and moment conditions. To give the theory a concrete footing—and to obtain closed-form bias constants in our finite-sample analysis—we instantiate this family with a data-adaptive *local-linear* construction, but all results hold for the broader class.

For every observable  $f$  we compute the one-step cumulant-generating function (CGF) and show that its quadratic coefficient  $\mathcal{I}_{\theta,\sigma}[f]$  serves simultaneously as (i) a *Fisher load*, (ii)  $\frac{2}{\theta^2}$  times the synthetic Ricci curvature  $\Gamma_2^{(\theta,\sigma)}(f)$ , and (iii)  $-2 \partial_{\sigma}$  of the one-step Shannon entropy. This *Fisher-curvature-entropy bridge* yields a single quadratic objective whose minimiser in the dial plane,  $(\theta^*, \sigma^*)$ —the *little-rose*—optimises curvature bias, entropy production, and operator-learning risk at once. We prove

$$\nabla_{(\theta,\sigma)} \text{KL}(\mu_1 \parallel k_{\theta,\sigma} \# \mu_0) = -\frac{1}{2} \nabla_{(\theta,\sigma)} \mathcal{I}_{\theta,\sigma},$$

so steepest KL descent converges to the little-rose with no hyper-parameters. A data-driven Goldenshluger–Lepski selector recovers  $(\hat{\theta}_{\star}, \hat{\sigma}_{\star})$  from finite samples, and we give non-asymptotic risk bounds for both Perron–Frobenius and Koopman estimates.

The resulting dial matrices  $(\Theta_{ij}, \Sigma_{ij})$  form a directed, doubly-weighted graph, opening avenues toward discrete curvature flows, entropy production on cycles, and multiparameter persistent homology—highlighting a promising intersection of statistical dynamics, information geometry, and topological data analysis.

## 1 Introduction

### 1.1 Why tune local kernels?

### 1.2 Key contributions

- Closed-form bridge: CGF curvature  $\rightarrow$  Fisher load  $\rightarrow$  curvature & entropy.
- Single KL optimisation (little-rose) jointly tunes **two** dials  $(\theta, \sigma)$ .
- Finite-sample Goldenshluger–Lepski selector with risk guarantees.

### 1.3 Organisation of the paper

## 2 Rose kernels and dials

### 2.1 Product-form kernels and minimal assumptions (K0–K3)

$$z := y - x, \quad r := \|z\|.$$

(W1) (**Radial weight**)  $w : [0, \infty) \rightarrow [0, \infty)$  is  $C^1$ , radial,  $\int_{\mathbb{R}^d} w(\|z\|) dz = 1$ ,  $\int \|z\|^2 w(\|z\|) dz < \infty$ , and  $w \in L^2(\mathbb{R}^d)$ .

(W2) (**Centred residual**)  $g : \mathbb{R}^d \rightarrow [0, \infty)$  is  $C^1$ ,  $\int g(z) dz = 1$ ,  $\int z g(z) dz = 0$ ,  $\int z z^\top g(z) dz < \infty$ , and  $\nabla g \in L^2(\mathbb{R}^d)$ .

**Definition 2.1** (Rose kernel). For  $(\theta, \sigma) \in (0, \infty)^2$  set

$$w_\theta(r) := \theta^{-d} w(r/\theta), \quad g_\sigma(z) := \sigma^{-d} g(z/\sigma), \quad k_{\theta,\sigma}(y | x) := \frac{w_\theta(\|x - y\|) g_\sigma(y - x)}{\int_X w_\theta(\|x - y'\|) g_\sigma(y' - x) dy'}$$

(K1) (**Markov**)  $k_{\theta,\sigma}(y | x) \geq 0$  and  $\int_X k_{\theta,\sigma}(y | x) dy = 1$ .

(K2) (**Score in  $L^2$** )  $S_{\theta,\sigma}(y | x) := \nabla_y \log k_{\theta,\sigma}(y | x) \in L^2(dy dx)$ .

(K3) (**Second-moment scaling**)  $\int_X \|y - x\|^2 k_{\theta,\sigma}(y | x) dy = O(\theta^2)$  as  $\theta \downarrow 0$ .

(K4) ( **$C^1$  in dials**)  $\partial_\theta \log k_{\theta,\sigma}, \partial_\sigma \log k_{\theta,\sigma} \in L^2(dy dx)$ .

### 2.2 Local-linear residual construction (canonical example)

### 2.3 Scaled kernels and notation for the rest of the paper

## 3 Population theory

(K0)–(K2) as in Sec. 3.0 — Markov, score  $L^2$ , second-moment scaling.

### 3.1 CGF $\Rightarrow$ Fisher $\Rightarrow$ Landau

**Definition 3.1** (One-step CGF).

**Proposition 3.1** (CGF–Fisher correspondence).

*Remark 3.1* (Legendre dual = Landau stiffness). A two-line box reminding readers that the Legendre transform of  $\Lambda$  yields a quadratic Landau free energy.

### 3.2 Fisher–curvature–entropy bridge

**Lemma 3.1** (Curvature identity).

**Lemma 3.2** (Discrete de Bruijn).

**Theorem 3.1** (Fisher–curvature–entropy bridge).

### 3.3 Introducing the dial plane

**(K0)  $C^1$  regularity in dials.** The maps  $(\theta, \sigma) \mapsto \log k_{\theta, \sigma}$  are  $C^1 \dots$

### 3.4 KL gradient and the *little-rose* optimiser

**Theorem 3.2** (KL gradient).

**Definition 3.2** (Little-rose).

**Corollary 3.1** (Unified optimality).

## 4 Finite-sample dial selection

### 4.1 Goldenshluger–Lepski selector

### 4.2 Risk bounds for PF and Koopman estimates

**Theorem 4.1** (GL selector, risk bound).

### 4.3 Illustrative experiment

## 5 Illustrative experiment: 2-D Langevin double-well

### 5.1 Model and data generation

- Potential:  $V(x, y) = \frac{1}{4} (x^2 - 1)^2 + \frac{1}{2} y^2$ .
- Over-damped Langevin SDE  $dX_t = -\nabla V(X_t) dt + \sqrt{2D} dW_t$ ,  $D = 0.04$ .
- Euler–Maruyama, step  $h = 0.01$ ;  $T = 2 \times 10^4$  steps. Store  $\mu_0 = \{X_{t_k}\}_k$  and  $\mu_1 = \{X_{t_k+h}\}_k$ .

### 5.2 Dial-plane scan

- Grid:  $\theta \in [0.05, 0.6]$  (25 pts),  $\sigma \in [0.02, 0.25]$  (25 pts).
- For each grid point compute  $\text{KL}(\mu_1 \parallel k_{\theta, \sigma} \# \mu_0)$  and  $\mathcal{I}_{\theta, \sigma}[x]$ .
- *Figure 5a–b*. Heat-maps of KL and Fisher; overlay location of minimum/maximum.

### 5.3 KL-gradient path (little-rose convergence)

- Initialise at  $(\theta_0, \sigma_0) = (0.5, 0.25)$ ; run projected gradient descent on KL (step 0.005).
- *Figure 5a* overlay white arrow showing path converging to  $(\theta^*, \sigma^*)$ .

## 5.4 Goldenshluger–Lepski selector

- 50 bootstrap splits (80/20); run GL rule over same grid.
- *Figure 5c.* Box-plot of selected  $(\hat{\theta}, \hat{\sigma})$  vs. oracle  $(\theta^*, \sigma^*)$ .

## 5.5 Operator-learning risk

- Estimate Koopman one-step map at three dial settings: GL, too-small  $\theta$ , too-large  $\sigma$ .
- Report MSE on held-out data. *Table 1.*

## 5.6 Landau free-energy curve

- At GL dials compute CGF for  $f(x, y) = x$ ; fit quadratic  $F[m] = \frac{1}{2}am^2$ .
- *Figure 5d.* Histogram of  $x$ -fluctuations with quadratic overlay; report  $a \simeq 1/\mathcal{I}$ .

## 5.7 Directed curvature teaser.

Discrete Ricci asymmetry on the dial graph is visualised in Appendix D.

# 6 Discussion and Outlook

## 6.1 Key implications

### (I1) Unified diagnostic of fluctuation strength

The Fisher load  $\mathcal{I}_{\theta, \sigma}$  is *simultaneously* the variance coefficient, the synthetic-curvature prefactor, and the entropy-production rate. A single dial pair  $(\theta, \sigma)$  governs them all.

### (I2) Automatic bias–variance balance

Steepest descent of  $\text{KL}(\mu_1 \parallel k_{\theta, \sigma} \# \mu_0)$  is steepest *ascent* of  $\mathcal{I}_{\theta, \sigma}$ , so the statistically most informative kernel is found with no cross-validation.

### (I3) One-step Schrödinger bridge

The little-rose kernel realises the discrete entropic transport  $\mu_0 \rightarrow \mu_1$ , linking operator learning to optimal-transport theory.

### (I4) Landau lens on data

The Legendre transform of the CGF equips any observable with a Landau-type free-energy landscape whose stiffness is data-computable.

## 6.2 Future directions

### (F1) Directed discrete curvature

Dial matrices  $(\Theta_{ij}, \Sigma_{ij})$  define a doubly-weighted digraph, opening analytic work on Ollivier/Bakry–Émery curvature for non-reversible kernels.

### (F2) Multiparameter persistent homology

The two weights furnish a natural bifiltration, inviting multi-parameter TDA of dynamical data sets.

**6.3** Limitations

**6.4** Conclusion

**A** Proof details for Sec. 3

**B** GL selector derivations

**C** Additional experiments

**D** Dial-graph topology (exploratory)